

4.1 Analytical Techniques

Question Paper

Course	CIEA Level Chemistry
Section	4. Analysis
Topic	4.1 Analytical Techniques
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Data from the mass spectrum of a sample of pure iron is shown in the table below.

m/e	Relative peak height
28	0.1
54	6.3
56	100.0
57	2.4
58	0.3

Which species is most likely to cause the peak at $m/e = 28$?

- A. $^{84}\text{Sr}^{3+}$
- B. $^{28}\text{Si}^+$
- C. $^{56}\text{Fe}^{2+}$
- D. $^{28}\text{Fe}^+$

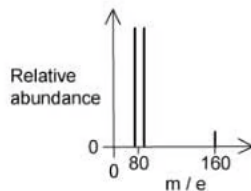
[1 mark]

Question 2

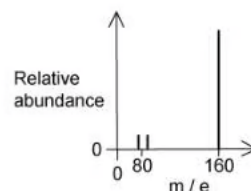
Two stable isotopes of bromine have relative masses of 79 and 81.

Which is the correct pattern of peaks in the mass spectrum of molecular bromine?

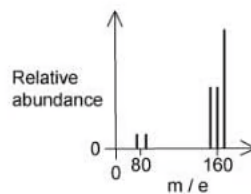
A.



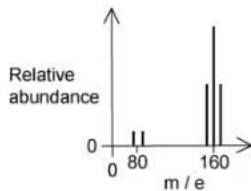
B.



C.



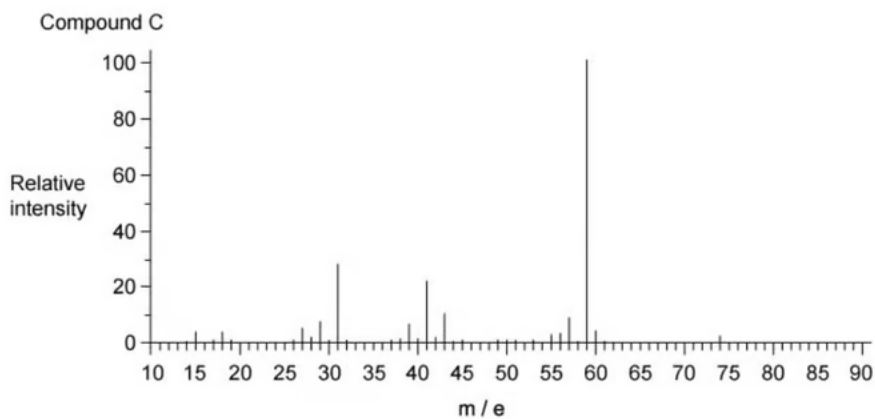
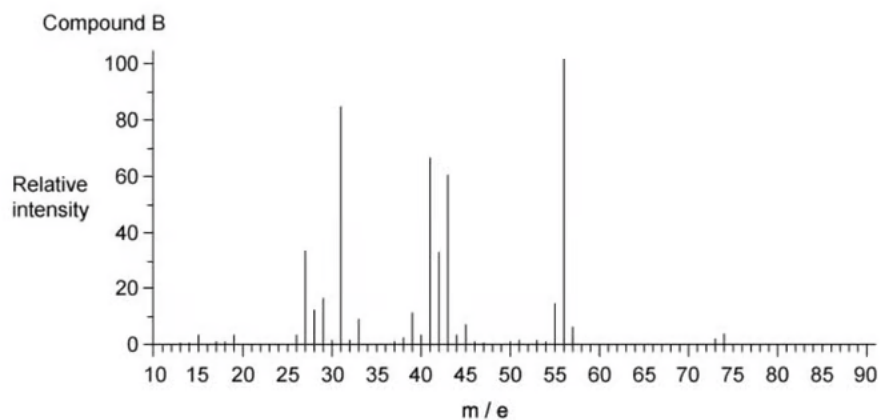
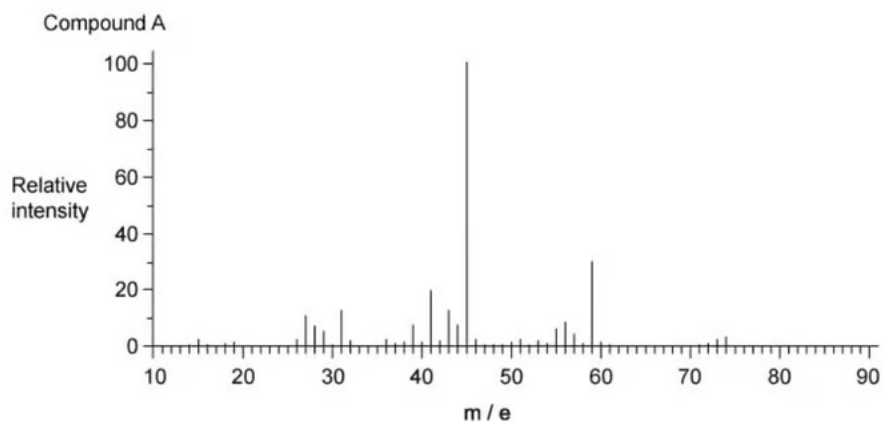
D.



[1 mark]

Question 3

The mass spectra of three compounds A, B and C are shown below.



Which of the following features of the three compounds A, B and C shows the molecular formula is $C_4H_{10}O$?

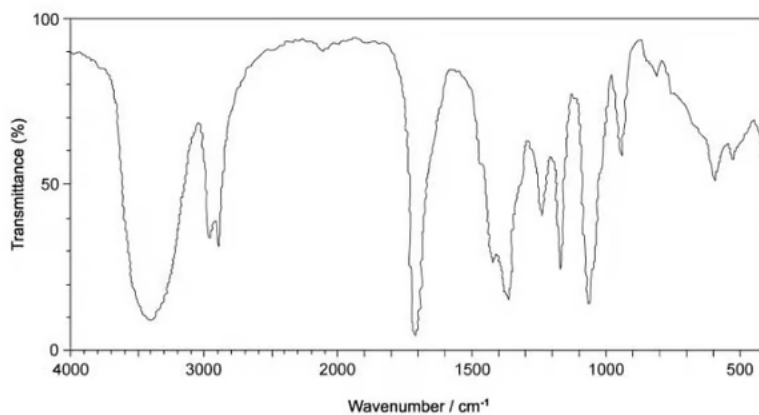
- A. All show a molecular ion peak at 74

- B. All show a molecular ion peak at 13
- C. All show a molecular ion peak at 73
- D. All show a molecular ion peak at 33

[1 mark]

Question 4

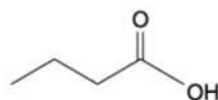
The infrared spectrum below shows an unknown compound.



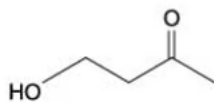
Bond	Wavenumber range / cm ⁻¹
C-O; hydroxyl, ester	1040 - 1300
C=C; aromatic compound, alkene	1500 - 1680
C=O amide	1640 - 1690
carbonyl, carboxyl	1670 - 1740
ester	1710 - 1750
C≡N; nitrile	2200 - 2250
C-H; alkane	2850 - 2950
N-H; amine, amide	3300 - 3500
O-H; carboxyl	2500 - 3000
hydroxyl	3200 - 3600

Which compound could have produced the infrared spectrum?

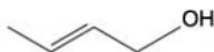
A.



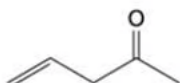
B.



C.



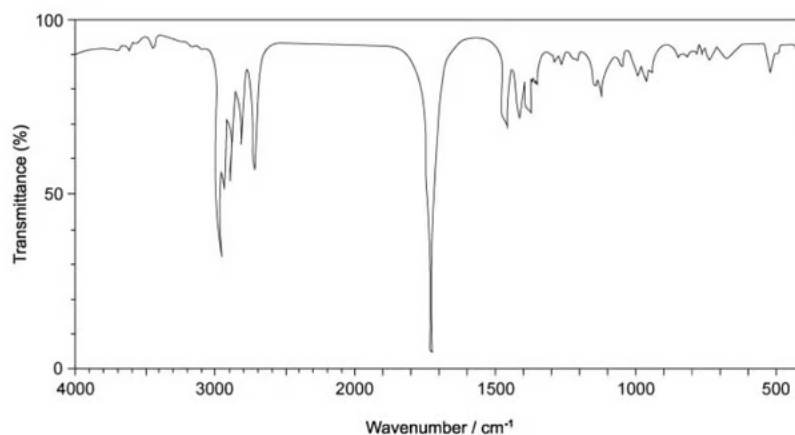
D.



[1 mark]

Question 5

The diagram shows an infrared spectrum of a compound.



Bond	Wavenumber range / cm ⁻¹
C–O; hydroxyl, ester	1040 – 1300
C=C; aromatic compound, alkene	1500 – 1680
C=O amide	1640 – 1690
carbonyl, carboxyl	1670 – 1740
ester	1710 – 1750
C≡N; nitrile	2200 – 2250
C–H; alkane	2850 – 2950
N–H; amine, amide	3300 – 3500
O–H; carboxyl	2500 – 3000
hydroxyl	3200 – 3600

Which compound would give this spectrum?

- A. Butanoic acid
- B. Butanal
- C. Butan-1-ol
- D. 1-bromobutane

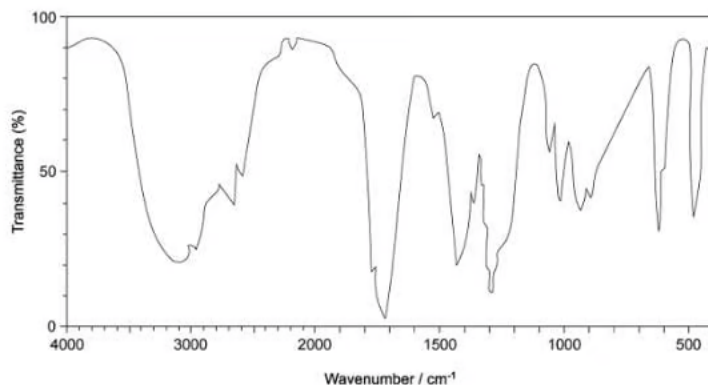
[1 mark]

Question 6

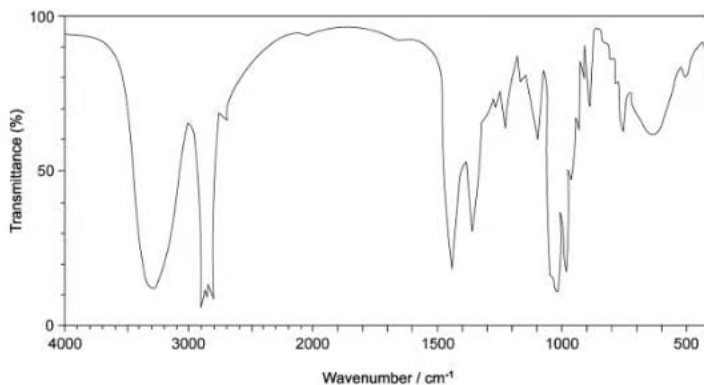
Which of the following infrared spectra could show a carboxylic acid?

Bond	Wavenumber range / cm^{-1}
C–O; hydroxyl, ester	1040 – 1300
C=C; aromatic compound, alkene	1500 – 1680
C=O amide	1640 – 1690
carbonyl, carboxyl	1670 – 1740
ester	1710 – 1750
C≡N; nitrile	2200 – 2250
C–H; alkane	2850 – 2950
N–H; amine, amide	3300 – 3500
O–H; carboxyl	2500 – 3000
hydroxyl	3200 – 3600

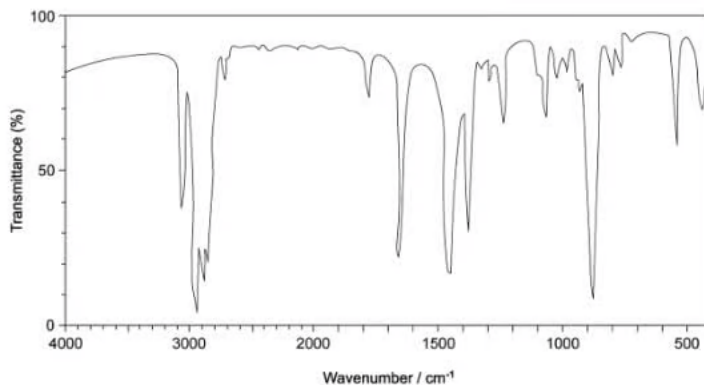
A.



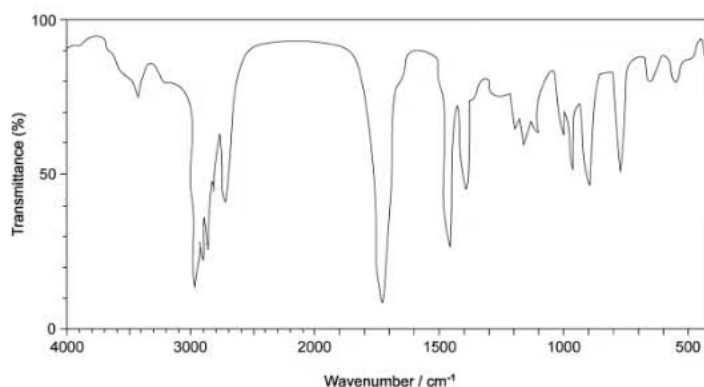
B.



C.



D.



[1 mark]

Question 7

Chlorine exists as two isotopes ^{35}Cl and ^{37}Cl in the ratio 3:1.

Which of the following statements about the molecular ion peaks in the mass spectrum of Cl_2 is correct?

- A. Peaks at $m/e = 70$ and 72 in the ratio 3:1
- B. Peaks at $m/e = 70$, 72 and 74 in the ratio 9:3:1
- C. Peaks at $m/e = 70$, 72 and 74 in the ratio 9:6:1
- D. Peaks at $m/e = 70$ and 74 in the ratio 3:1

[1 mark]

Question 8

A compound XF_n is a fluoride of another halogen, X, and it is known that $n > 1$. The highest mass peak in the spectrum of XF_n is assigned to the parent ion and comes as single peak at $m/e = 222$.

Which of the following statements is **incorrect**?

- A. $n = 5$
- B. The compound could contain bromine
- C. X and F are monotopic
- D. The compound is a fluoride of iodine

[1 mark]

Question 9

In the mass spectrum of trifluoroacetic acid, $\text{CF}_3\text{CO}_2\text{H}$, intense peaks are observed at $m/e = 69$ and 45 (base peak) in addition to other peaks.

The peak at $m/e = 69$ is accompanied by a peak at $m/e = 70$, which is about 1.1% the intensity of the peak at $m/e = 69$.

Which of the following statements is not supported by this information?

- A. C–C bond cleavage occurs
- B. Fluorine is monotopic
- C. $[\text{HO}_2\text{C}]^+$ is a fragment ion
- D. $\text{CF}_3\text{CO}_2\text{H}$ fragments by sequential loss of F atoms

[1 mark]

Question 10

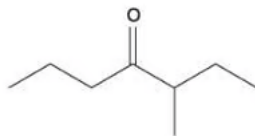
A sample is analysed by mass spectrometry and infrared spectroscopy. The following incomplete data is obtained.

Mass spectrometry	Infrared spectroscopy
Molecular ion peak at $m/e = 128$	Sharp peak at 1720 cm^{-1}
Fragment ion peak at $m/e = 15$	No broad peak around 3200 cm^{-1}

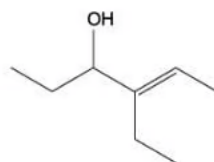
Which of the following compounds is demonstrated by this data?

Bond	Wavenumber range / cm^{-1}
C–O; hydroxyl, ester	1040 – 1300
C=C; aromatic compound, alkene	1500 – 1680
C=O amide	1640 – 1690
carbonyl, carboxyl	1670 – 1740
ester	1710 – 1750
C≡N; nitrile	2200 – 2250
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O–H; carboxyl	2500 – 3000
hydroxyl	3200 – 3600

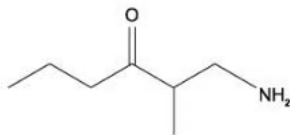
A.



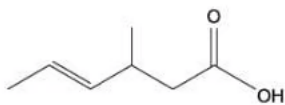
B.



C.



D.



[1 mark]

Model Answers: Hard

1

The correct answer is **C** because:

- The m/e value is a mass to charge ratio; this represents the mass divided by the charge number.
- The mass of iron is 56, and the charge is +2, so this will show on a mass spectrum at the peak of 28.

A & B are incorrect as the sample is of pure iron so no other elements would be present.

D is incorrect as this is the incorrect mass for iron.

2

The correct answer is **D** because:

- Bromine is a diatomic molecule there will be 5 peaks on the mass spectrum of bromine.
- Bromine consists of molecules, not individual atoms. When bromine is passed through the mass spectrometer, an electron is given off to give the **molecular ion, Br_2^+** .
- Some of these will fragment to make $\text{Br} + \text{Br}^+$
 - $\text{Br}_2^+ \rightarrow \text{Br} + \text{Br}^+$
- The Br atom passes through the machine, and the Br^+ ions will give lines at 79 and 81.
- There will also be a line for the unfragmented Br_2^+ ion.
- This will give 3 molecular ion peaks.
 - Br_2^+ ion containing the isotopes $79 + 79 = 158$
 - Br_2^+ containing the isotopes $79 + 81 = 160$

- Br_2^+ containing the isotopes $81 + 81 = 162$

A is incorrect as this only shows one molecular ion peak.

B is incorrect as this only shows one molecular ion peak.

C is incorrect as the graph shows three molecular ion peaks with wrong relative abundances.

3

The correct answer is **A** because:

- The peak representing the **molecular ion (M^+ or M_e^+)** is called the **molecular ion peak**.
- Excluding any peaks due to the presence of heavier isotopes, this is the peak with the highest m/e ratio.
- In all three graphs, the molecular ion peak is at 74.

B, C & D are incorrect as these are not the heaviest peaks in all the graphs.

4

The correct answer is **B** because:

- Infrared is the analysis of infrared light interacting with a molecule; this can be measured by absorption, emission and reflection.
- It is used to determine functional groups in atoms based on the bond strength and mass of the atoms.
- The presence of a peak around 1670–1740 indicates that the compound contains a carbonyl or carboxyl group.
- The O–H peak matches the alcohol at 3200 – 3600; a carboxyl OH group would be lower at 2500–3000.
- The compound cannot be a carboxylic acid, but it contains separate OH and C=O groups.

A is incorrect as the band for the O–H peak would have been much lower for the carboxylic acid group, range of 2500 – 3000.

C is incorrect as there is no characteristic peak at 1500–1680 for a C=C and the spectrum has a C=O which is missing in the molecule.

D is incorrect as there is no characteristic peak at 1500–1680 for a C=C and the spectrum has an OH group which is absent in this molecule.

Exam tip: make sure you label the wavelengths with the correct bond; a common misconception is to label the C–H peak with O–H.

5

The correct answer is **B** because:

- There is a strong peak at 1710 – 1750, this shows a C=O bond in an ester or a carbonyl or carboxyl.

carbonyl or carboxyl group.

- The O-H peak matches the alcohol at 3200 – 3600; a carboxyl OH group would be lower at 2500–3000.
- The compound cannot be a carboxylic acid, but it contains separate OH and C=O groups.

A is incorrect as the band for the O–H peak would have been much lower for the carboxylic acid group, range of 2500 – 3000.

C is incorrect as there is no characteristic peak at 1500–1680 for a C=C and the spectrum has a C=O which is missing in the molecule.

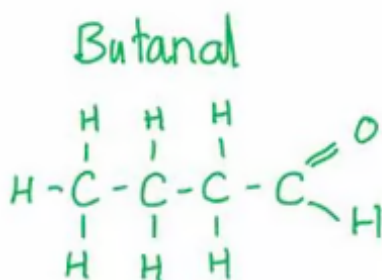
D is incorrect as there is no characteristic peak at 1500–1680 for a C=C and the spectrum has an OH group which is absent in this molecule.

Exam tip: make sure you label the wavelengths with the correct bond; a common misconception is to label the C–H peak with O–H.

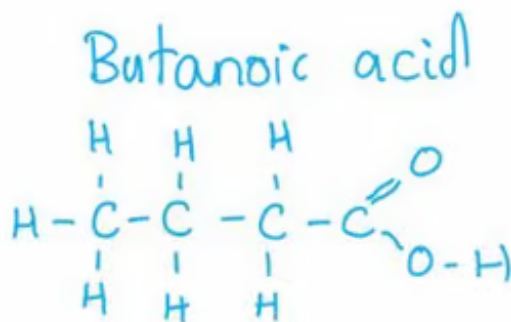
5

The correct answer is **B** because:

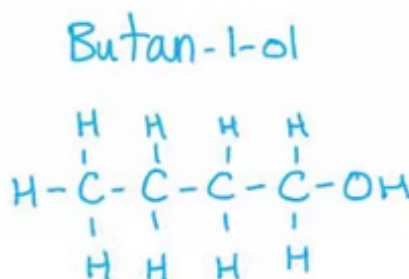
- There is a strong peak at 1710 – 1750, this shows a C=O bond in an ester or a carbonyl or carboxyl.
- There is a wider peak from 2850 – 2950 this shows a C–H bond in alkanes.



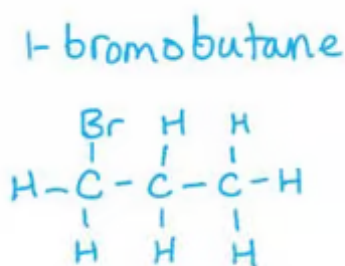
A is incorrect as butanoic acid would have a peak at 1670 – 1740 for the carboxyl group. The peak around 3000 would be much broader due to the presence of the carboxyl group and the O–H bond and the C=O bond.



C is incorrect butan-1-ol would have a much more defined peak at 3200 – 3600 due to the hydroxyl group. There would be no peak at 1700 as there is no C=O bond.



D is incorrect as 1-bromobutane there would be no peak at 1700 as there is no C=O bond.



6

The correct answer is **A** because:

- In a carboxylic acid the peak around 3000 would be much broader due to the presence of the carboxyl group and the O–H bond and the C=O bond.

B is incorrect as this is an IR spectrum of an alcohol, seen by the broad absorption at 3200 – 3600 which is characteristic of the O–H in the alcohol.

C is incorrect as this is an IR spectrum of ethanoate showing a sharp peak between 1710 and 1750 showing the carbonyl group in the ethanoate.

D is incorrect as this is an IR spectrum of a ketone due to sharp absorbance at 1710 characteristic of the $\text{C}=\text{O}$ bond in the ketone.

7

The correct answer is **C** because:

- Chlorine consists of molecules, not individual atoms. When chlorine is passed through the mass spectrometer, an electron is given off to give the **molecular ion**, Cl_2^+ .
- Some of these will fragment to make Cl^+ and Cl^-
 - $\text{Cl}_2^+ \rightarrow \text{Cl}^+ + \text{Cl}^-$
- The Cl^+ atom passes through the machine and the Cl^- ions will give lines at 35 and

presence of the carboxyl group and the O–H bond and the C=O bond.

B is incorrect as this is an IR spectrum of an alcohol, seen by the broad absorption at 3200 – 3600 which is characteristic of the O–H in the alcohol.

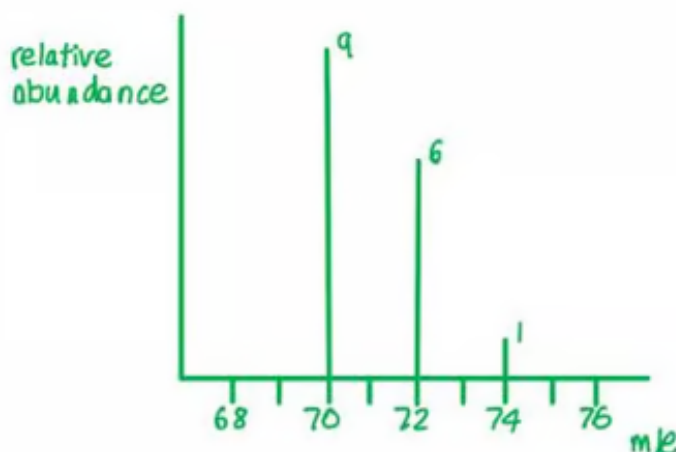
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D is incorrect as this is an IR spectrum of a ketone due to sharp absorbance at 1710 characteristic of the C=O bond in the ketone.

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The correct answer is **C** because:

- Chlorine consists of molecules, not individual atoms. When chlorine is passed through the mass spectrometer, an electron is given off to give the **molecular ion**, Cl_2^+ .
- Some of these will fragment to make Cl^+ and Cl^+
 - $\text{Cl}_2^+ \rightarrow \text{Cl} + \text{Cl}^+$
- The Cl atom passes through the machine and the Cl^+ ions will give lines at 35 and 37.
- There will also be a line for the unfragmented Cl_2^+ ion.
- This will give 3 molecular ion peaks
 - Cl_2^+ ion containing the isotopes $35 + 35 = 70$
 - Cl_2^+ containing the isotopes $35 + 37 = 72$
 - Cl_2^+ containing the isotopes $37 + 37 = 74$



8

The correct answer is **B** because:

- The relative atomic mass of bromine is 79.9 if this is taken away from the mass in the spectrum of 222 it would leave a mass of 142.1
- A mass of 142.1 divided by the RAM of fluorine would mean that the compound would have 7.47 atoms of fluorine.

A is incorrect as the RAM of iodine is 127 if this is taken away from 222 it leaves 95, 95 divided by 19 (RAM fluorine) is 5. Therefore, $n = 5$.

C is incorrect as both fluorine and iodine are of the 26 elements that have a single stable isotope.

D is incorrect as iodine fits all the other correct statements as the halogen bonded with the fluorine, so this is correct.

9

The correct answer is **D** because:

- There will not be sequential loss of F atoms as this would not account for the peaks at 45 and 69 which are the two fragments produced from this molecule.
- The peak at 70 would be explained by isotopes of carbon being present in the

molecule as fluorine is monatomic.

A is incorrect as the molecule will fragment around the C–C bond.

B is incorrect as fluorine is one of the 26 elements that have a single stable isotope.

C is incorrect as one of the fragments will be the $[\text{HO}_2\text{C}]^+$ ion.

10

The correct answer is **A** because:

- A sharp peak at 1720 cm^{-1} is explained by the $\text{C}=\text{O}$ in molecule **A**.
- The molecular ion peak of 128 would be explained by molecule **A**.
- The fragment ion peak of m/e 15 would be explained by a methyl ion.

B is incorrect as it has been ruled out using the IR data, there is no peak for the hydroxyl